# Aswin Sanosh

PROGRAMMER & 3D ARTIST

## Contact


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India, 683516.


## Links

https://github.com/AswinSanosh

https://www.linkedin.com/in/aswinsanosh/

## Languages

- English
- Malayalam

## Soft Skills

- Problem Solving
- Critical Thinking
- Leadership
- Technical Mentoring
- Team Collaboration
- Innovation Management
- Communication
- Adaptability
- Analytical Thinking
- Continuous Learning

## Personal Details

- Passport Number: V2383434
- Date of Birth: 24-September-2002
- Nationality: India

## Career Objective

To leverage my expertise in full-stack development, AI/ML, and IoT within a research-driven academic or innovation-focused environment, where I can contribute to cutting-edge projects, deepen my technical knowledge, and create scalable solutions that bridge software, hardware, and real-world impact.

## Education

**Master of Computer Applications** 2021 - 2026
*SAINTGITS COLLEGE OF ENGINEERING*
Kottayam, Kerala, India
CGPA: 7.54/10

**Bachelor of Science (BSCM)** 2024 - 2027
*INDIRA GANDHI NATIONAL OPEN UNIVERSITY*
Ernakulam, Kerala, India
CGPA: N/A

**Plus Two (Senior Secondary)** 2020 - 2021
*BHARATIYA VIDYA BHAVANS VIDYA MANDIR*
Thrissur, Kerala, India
Score: 80%

**Tenth Standard (Higher Secondary)** 2018 - 2019
*MOUNT CARMEL VIDHYANIKETAN*
Kottayam, Kerala, India
CGPA: 91.4%

## Experience

**Intern** Dec 2025 – present
*NATIONAL CHUNG CHENG UNIVERSITY*
Minxiong, Chiayi, Taiwan
- Digital twin design for flood inundation using physics-informed neural networks

**Teacher (Part Time)** Feb 2024 – Dec 2025
*GURUKULAM TUITION CENTER*
Kottayam, Kerala, India
- Teaching high-school students on Physics, Chemistry, Mathermatics, Computer Science and Social Science.

**Intern** July 2024 – Sep 2024
*NATIONAL CHUNG CHENG UNIVERSITY*
Minxiong, Chiayi, Taiwan
- Full-Stack Development of Stock Prediction Website using Django and React.js

**Software Mentor and Developer** Feb 2025 – present
*TESSAT SPACE PVT. LTD. (STARTUP)*
Kottayam, Kerala, India
- Mentoring software development team of Tessat Space.
- Developing Softwares (Full Stack) and research in AI/ML projects

**Web-Master** Dec 2024 – present
*IEEE COMPUTER SOCIETY STUDENT CHAPTER*
Saintgits College of Engineering
Kottayam, Kerala, India

- Collaborate with the team to implement web-based solutions for events, announcements, and member engagement.

**Chief Marketing Officer** July 2023 – Feb 2024
*IEDC@SAINTGITS*
Saintgits College of Engineering, Kottayam, Kerala, India

- Mentoring software development team of Tessat Space.
- Developing Softwares (Full Stack) and research in AI/ML projects

## Technical Skills

### Programming Languages
Python, JavaScript, Java, C++, C, C#, SQL, R, PHP, Dart, VB.NET, HTML/CSS

### Frameworks & Libraries
- Django, Django REST Framework, React.js, Next.js, Node.js, NestJS, Express.js, Flutter, React Native, Redux, jQuery, Three.js, Tailwind CSS, Material UI, GSAP, Framer Motion, Laravel PhP
- NumPy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn, TensorFlow, Keras, PyTorch, TorchVision, FastAI, OpenCV, MediaPipe, YOLOv8, OpenPose, UMAP, HDBSCAN, Hugging Face Transformers, Streamlit, TensorFlow Serving, FastAPI, Flask, MLflow

### Databases & Cloud
PostgreSQL, MySQL, MongoDB, SQLite, Firebase

### Tools & Platforms
Git, VS Code, PyCharm, Google Colab, Figma, Blender, Unity, Unreal Engine, Arduino IDE, Bambu Studio, Ultimaker Cura, Meta SparkAR

### Domains & Specializations
- Full-Stack Web Development
- AI/ML (Unsupervised Learning, Computer Vision, Pattern Recognition)
- IoT & Embedded Systems (Arduino, NFC, GPS)
- 3D Modeling, Printing & Web-based 3D Visualization
- Real-Time Data Systems & REST APIs

## Projects

**Stock Prediction Website (Full Stack)**
*React.js, Django, PostgreSQL, REST API*

- Developed a stock prediction platform with interactive graphs, wishlist tracking, and Head & Shoulders pattern detection. Integrated Yahoo Finance for market insights, implemented a Virtual Queue System for logins, membership plans, and real-time alerts, delivering a seamless trading experience.
- **Application:** Stock trend forecasting and market analysis.

**Digital Twin Design for Flood Inundation (Physics-Informed AI)**
*Python, PyTorch, Graph Neural Networks, HEC-RAS*

- Developed a physics-informed digital twin for real-time flood inundation forecasting using graph neural networks on unstructured meshes. Integrated hydrodynamic simulation outputs (HEC-RAS) with terrain and rainfall data, enforcing mass conservation and physical constraints through custom loss functions. Enabled fast, stable flood prediction with interpretable model behavior, supporting scenario analysis, early warning, and decision-making for extreme flood events.
- **Application:** Real-time flood forecasting, risk assessment, and disaster management.

**Body Shape Classification Using OpenCV, OpenPose, and Shapy AI**
*Machine Learning, CUDA, Python, OpenCV, OpenPose*

- Developed AI models to classify human body shapes (rectangle, pear, apple, hourglass, etc.) using pose estimation and anthropometric analysis. Integrated OpenPose and Shapy AI for accurate 3D body geometry extraction and visual measurement-based prediction.
- **Application:** Personalized fashion recommendations, virtual fitting, and health analytics.
- **Link:** https://github.com/AswinSanosh/chic2

**Unsupervised Learning for GPR Image Classification**
*Python, Keras, K-means, BIRCH, DBSCAN, HDBSCAN, DINO-V3*

- Designed unsupervised clustering pipelines using K-Means, DBSCAN, and BIRCH to classify road distress patterns from GPR imagery. Integrated feature extraction using ResNet, MobileNet V2, DINO V3 and data augmentation for enhanced accuracy.
- **Application:** Road condition monitoring and subsurface damage detection.
- **Link:** https://github.com/AswinSanosh/GPR-ANALYSIS

**Classification of Industrial Tools using traditional methods**
*C++, Lua, OpenCV, Hu-moments, HOG, SVG*

- Developed a visual classification model for industrial tools (e.g., screws, gears, and equipment) using HOG-SVG descriptors without traditional ML frameworks. Designed for real-time object detection in manufacturing environments.
- **Application:** Automated tool recognition in industrial inspection systems and comparison of performance between Machine Learning and traditional methods.
- **Link:** https://github.com/AswinSanosh/HOG-SVG

**GPS-Integrated Bus Schedule Website**
*Next.js, IoT, GPS, NFC*

- Developed a real-time bus tracking and ticketing platform integrating GPS and NFC technology for cashless travel. Enabled contactless payments and live route visualization for commuters.
- **Application:** Smart public transport and digital ticketing systems.
- **Link:** https://github.com/AswinSanosh/bus_tracker

**Samyuktha 2025 Website**
*Next.js, Redux, Tailwind CSS, MySQL*

- Created the official web platform for Saintgits College's annual techno-cultural fest. Included real-time event updates, workshop registrations, announcements, and user dashboards for participants.
- **Application:** Event management, Registrations, Payments and digital festival coordination.
- **Link:** https://samyukta2k25.vercel.app

**Tessat Space Pvt. Ltd. Flight Software (CANSAT)**
Django, React.js, MySQL

- Engineered real-time flight software for satellite data visualization and anomaly detection. Designed telemetry dashboards and alert systems to support satellite operations.
- **Application:** Satellite monitoring and data analytics.
- **Link:** https://github.com/AswinSanosh/CANSAT

**Helekin Technologies Website**
*Next.js, MySQL, Three.js*

- Developed an interactive 3D model customization and slicing platform for 3D printing services. Integrated secure payments, order tracking, and an admin dashboard for product management.
- **Application:** 3D printing e-commerce and online model customization.
- **Link:** https://www.helekin.in/

**Redwills Interactive Website**
*Next.js, Three.js, WebGL*

- Developed a visually immersive and performance-optimized website showcasing the studio's expertise in 3D game development. The platform highlights their flagship project, studio portfolio, and brand identity with interactive 3D elements and smooth animations, delivering a strong narrative-driven experience aligned with gaming aesthetics.

**Application:** Game studio portfolio, IP showcase, and brand presence for a 3D game development company.
**Notable Project:** Thekku Island – Kerala's first 3D horror game.
**Link:** https://www.redwillsinteractive.com/

## Certificates / Awards

- Developing Applications with SQL, Databases, and Django - Coursera
- Python for Machine Learning & Data Science Masterclass - Udemy Crash Course
- Electronics and PCB Design - Udemy
- Won Facebook Meta SparkAR Competition - April 2021

## References

**Prof. Pao-Ann Hsuing**
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*National Chung Cheng University, Chiayi, Taiwan*
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